

Maggie Shao

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

M.S. in Machine Learning (Robotics Focus) GPA: 3.9/4.0

2025.8-present

Tsinghua University

Beijing, China

Bachelor of Mathematics, Physics and Industrial Engineering GPA: 3.8/4.0

2021.9-2025.6

University of California, Berkeley

Berkeley, CA

Exchange Student

2023.9-2024.1

EXPERIENCE

Research Intern — Siemens Research (Deep Learning Group)

Siemens Research, Munich and Beijing

2024.11-2025.1

- Contributed to the core development of General Time Transformer (GTT), an encoder-only foundation model for zero-shot multivariate time series forecasting, using curve-shape tokenization and channel-wise attention on large-scale Siemens industrial time series
- Pretrained GTT on a large-scale corpus totaling 2.9TB of diverse time series data, outperforming state-of-the-art forecasting models on multiple benchmarks
- Individual Contribution:** Led large-scale pretraining on a 2.9TB time series corpus, including 2TB of data that I independently sourced and processed, through direct outreach to multiple Siemens departments

PROJECTS

Weco: Equivariant World Changes for Object-Centric Policy Learning

2026.5-present

Collaborated with Yuejiang Liu, advised by Prof. Chelsea Finn, Stanford University

- World-Model-Based Planning
- Position-Equivariant, GeometryInvariant IDM
- Leveraging Large-Scale 2D Visual Priors given limited limited robot demonstrations
- Generalize across manipulation skills, objects, backgrounds, spatial configurations, and robot embodiments

(CVPR VLA) AbsVLA: Robust Vision-Language-Action Policies via Object-Centric Abstraction

2025.9-present

Advised by Prof. Zackory, Robotics Institute, Carnegie Mellon University

- Proposed [AbsVLA](#), a framework that trains robot policies in an object-centric abstract state space to improve robustness of VLA models; Fine-tuned OpenVLA-7B with LoRA on 200k demonstrations for parameter-efficient multi-task learning; Integrated vision – language grounding (Grounded-SAM) with manipulation policies and mapped instructions to 7 primitive skills.
- Achieved stronger robustness on LIBERO under visual and language distribution shifts; Demonstrated zero-shot sim-to-real transfer from Franka Panda simulation to Hello Robot Stretch.

From-Scratch Deep Learning Systems: DNN, CNN, Transformer & Diffusion

Intermediate Deep Learning Course project (Prof. Ruslan Salakhutdinov), CMU

2025.8-Present

- Implemented core neural networks entirely **from scratch** in Python — including linear layers, dropout, momentum optimizer, full backprop; additionally built CNNs, Transformers, and Diffusion models from scratch

- Trained and evaluated: (1) DNN (Fashion-MNIST classification); (2) CNN (Tiny-ImageNet 15-class vision); (3)Transformer (image-to-text description generation); and (4) diffusion models (generate clothing images from random Gaussian noise)

Design a skeleton CPU

2023.10-2023.12

Computer Architecture(CS 61C) Course project, University of California, Berkeley

- Designed a skeleton CPU that can execute the RISC-V instructions and pipeline

NLP Research Intern — Lenovo Research

2024.8-2024.10

AI Lab, Lenovo Research

- Researched architecture and development strategies for Copilot products designs leveraging novel platforms like Copilot, Palantir, and GPT-Engineer
- Designed evaluation frameworks and principles to guide the development of AI Copilot applications tailored to specific scenarios, prioritizing key aspects of functionality where existing capabilities were insufficient to meet all requirements
- **Individual Contribution:** Built [a Functional Demo](#) of Coding Copilot platform, serving as the foundation for downstream product and frontend integration.

Imitation Learning for Continuous Control

2025.8-2025.9

Robotics Institute, Carnegie Mellon University

- Implemented behavioral cloning (BC) and the DAgger algorithm for policy learning on continuous control tasks in OpenAI Gym (e.g., Ant-v2, Hopper-v2)

Conducted experiments comparing expert policies, BC, and DAgger; Tuned hyperparameters (learning rate, network depth, training data size), producing performance curves

Dynamic Intra-Cell Repositioning in Free-Floating Bike-Sharing Systems

Institute of Operations Research & Statistics, Tsinghua University

2024.8-Present

- Developed a Markov Decision Process (MDP) model and Policy Function Approximation (PFA) algorithm to optimize bike repositioning within urban cells, significantly reducing user wait times; Applied Approximate Dynamic Programming (ADP) and Optimal Computing Budget Allocation (OCBA) techniques, achieving higher bike availability at demand points
- Validated model through extensive simulations with real-world data from a major Chinese bike-sharing company, outperforming benchmark methods

Interactive 3D Visualization to enable Visual Analytics for Measurement Science

University of California, Berkeley; U.S. National Institute of Standards and Technology

2023.8-2023.12

- Developed an immersive data visualization platform using **Trame** and **ParaView**, integrating 2D and 3D views for interactive analysis in CAVETM and Head Mounted Displays
- Enabled dynamic data exploration by synchronizing and linking 2D and 3D visualizations with brushing techniques; Enhanced user interactivity through real-time data updates driven by selections across 2D or 3D visualizations

SKILLS

Machine Learning & Deep Learning: CNNs, RNNs, Transformers, Diffusion Models, GANs, VAEs, Attention, Reinforcement Learning (PPO, etc.), Multimodal Learning, Dropout, Momentum, Backpropagation, Hugging Face

Programming & Tools: **Computer Architecture**, Python, R, C/C++, SQL, MATLAB, Git, Unity, SolidWorks,

Relevant Projects: Second-hand Goods Trading Platform(SQL, Python, Full-Stack Development),

Library Seat Reservation Software(C++)

Computer Vision: Vision–Language Integration, Cross-Attention, Object Detection & Segmentation, Multimodal Perception, Representation Learning

Robotics: Kinematics, Policy Gradient, Imitation & Demonstration Learning(Behavior Cloning, Diffusion/DiT), Cross Attention, Vision–Language Models (VLMs), Masked Representations, Semantic Segmentation, Multimodal Learning, Simulation(OpenAI Gym, MuJoCo)

Cloud & Infrastructure: AWS (EC2/S3), Google Cloud, Docker

Mathematics: Calculus, Linear Algebra, Probability, Statistics, Statistical Inference, Stochastics